



YEAR 12 Mathematics Ext 1
HSC Course
Assessment Task 1
December 2009

1. There are 4 sections.
2. Answer each question on your own paper showing all necessary working
3. Start each section on a new page
4. Calculators may be used

Topic	Mark
1. Section 1 (Parametric Equations)	/15
2. Section 2 (Mathematical Induction)	/11
3. Section 3 (Trigonometry)	/11
4. Section 4 (Harder Trigonometry)	/7

TOTAL /44

Start a new page for each section.

Section 1. Parametric Equations (15 marks)

- i. Find the Cartesian equation of the following parabolas

a) $x = t$ (1 mark)

$$y = \frac{t^2}{2}$$

b) $x = t - 4$ (2 marks)

$$y = 2 + 2t + t^2$$

- ii. Write down the parametric equations (in terms of t) for the parabola $x^2 - 8y = 0$ (2 marks)

- iii. $P(4p, 2p^2)$ and $Q(4q, 2q^2)$ are two points on a parabola.

a) Show that the gradient of the chord PQ is $\frac{p+q}{2}$ (2 marks)

b) Hence show the equation of chord PQ is $y = \frac{1}{2}(p+q)x - 2pq$ (2 marks)

c) Show that if PQ is a focal chord then $pq = -1$ (1 mark)

- iv. The point $T(16t, 8t^2)$ lies on the parabola $x^2 = by$

a) Find the value of b (1 mark)

b) Show the equation of the tangent at T is given by $y = tx - 8t^2$ (2 marks)

c) The tangent at T cuts the y axis at $(0, -128)$ (2 marks)

What are the co-ordinates of T?

Section 2. Mathematical Induction (11 marks)

- i. Prove by Mathematical induction

a) $9^n - 3$ is divisible by 6 for all positive integers n (5 marks)

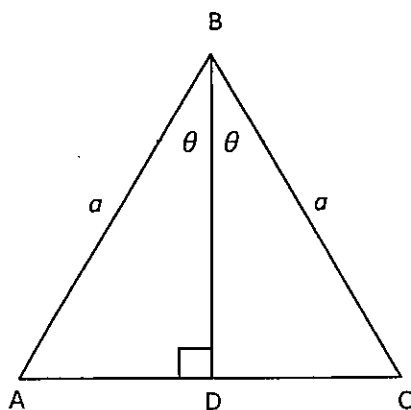
b) $2^n > 2n + 1$, $n \geq 3$ (6 marks)

Section 3. Trigonometry (11 marks)

- i. Given $f(x) = \sin^3 x + \cos^3 x$
Show that when $f(x) = 0$, $\tan x = -1$ (2 marks)
- ii. Write $\cos \theta - \sqrt{3} \sin \theta$ in the form $R \cos(\theta + \alpha)$ where α is in degrees
Hence solve $\cos \theta - \sqrt{3} \sin \theta = 1$ for $0^\circ \leq \theta \leq 360^\circ$ (5 marks)
- iii. Solve by the t method, where $t = \tan \frac{\theta}{2}$
 $3 \sin \theta - 4 \cos \theta = 4$ for $0^\circ \leq \theta \leq 360^\circ$
(Give solutions to θ to the nearest degree) (4 marks)

Section 4. Harder Trigonometry (7 marks)

- i. The triangle ABC is isosceles with $AB = BC = a$, and BD is perpendicular to AC.
Let $\angle ABD = \angle CBD = \theta$



- a) By considering the **area** of $\triangle ABD$ **only**, show that its area is $\frac{a^2 \sin \theta \cos \theta}{2}$ (2 marks)
- b) By considering the **area** of $\triangle ABC$, prove that $\sin 2\theta = 2 \sin \theta \cos \theta$ (2 marks)
- ii. Prove that $\frac{\sin 5\theta}{\sin \theta} - \frac{\cos 5\theta}{\cos \theta} = 4 \cos 2\theta$ (3 marks)

END OF EXAMINATION

Q1/

(i)(a) $y = \frac{x^2}{2}$

$x^2 = 2y$

(b) $y = 2 + 2(x+t) + (x+t)^2$ $x = t - 4$
 $x + 4 = t$

$y = 2 + 2x + 8 + x^2 + 8x + 16$

$y = x^2 + 10x + 26$

(ii) $x^2 = 8y$ $a = 2$ using $(2at, at^2)$

$(4t, 2t^2)$

$x = 4t$

$y = 2t^2$

(iii) $\frac{2p^2 - 2q^2}{4p - 4q} = \frac{2(p-q)(p+q)}{4(p-q)}$
 $= \frac{p+q}{2}$

(c) $y - 2p^2 = \frac{p+q}{2} (x - 4p)$

$y - 2p^2 = \frac{1}{2} (p+q)x - 2p^2 - 2pq$

$y = \frac{1}{2} (p+q)x - 2pq$

(d) Sub $(0, 2)$ $2 = -2pq$
 $-1 = pq$ (as required)

(iv) (a) $256t^2 = 8bt^2$
 $8b = 256$
 $b = 32$

(b) $x^2 = 32y$

$y = \frac{x^2}{32}$

$\frac{dy}{dx} = \frac{2x}{32}$

$= \frac{x}{16}$

at $x = 16t$

$m_T = t$

$y - 8t^2 = t(x - 16t)$

$y = tx - 16t^2 + 8t^2$

$y = tx - 8t^2$

(c) $-128 = 0 - 8t^2$

$t^2 = 16$

$t = \pm 4$

$\therefore T$ is $(64, 128)$
 $(-64, 128)$

Question 2

Step 1 Prove true for $n=1$

$$\text{ie } 9^1 - 3 = 6$$

①

$\therefore$ true for $n=1$

Step 2 Assume true for $n=k$ where $k \in \mathbb{N}$

$$\text{ie } 9^k - 3 = 6M \quad \text{where } M \in \mathbb{N}$$

$$9^k = 6M + 3$$

① $\frac{1}{2}$

Step 3 Prove true for $n=k+1$

ie $9^{k+1} - 3$ is divisible by 6

$$\begin{aligned} 9^{k+1} - 3 &= 9 \cdot 9^k - 3 \\ &= 9(6M + 3) - 3 \\ &= 54M + 27 - 3 \\ &= 54M + 24 \\ &= 6(9M + 4) \end{aligned}$$

$\therefore$ true for $n=k+1$

② $\frac{1}{2}$

Step 4 As it is true for $n=1$, and if true for $n=k$, it is true for $n=k+1$.

$\therefore$ true for all positive integers of n .

(b) Step 1 Prove true for $n=3$

$$2^3 > 2(3) + 1$$

$$8 > 7$$

$\therefore$ true for $n=3$

Step 2 Assume true for $n=k$, where k is a positive integer greater than or equal to 3

$$\text{ie } 2^k > 2k + 1$$

Step 3 Prove true for $n=k+1$

$$\begin{aligned} \text{ie } 2^{k+1} &> 2(k+1) + 1 \\ 2^{k+1} &> 2k + 3 \end{aligned}$$

$$2^k \cdot 2 > 2k + 3$$

$$(2k+1) \cdot 2 > 2k + 3$$

from assumption

$$4k + 2 > 2k + 3$$

As $k \geq 3$ the smallest difference that can occur for $4k - 2k$ is 6.

$$\therefore 4k + 2 > 2k + 3$$

$\therefore$ true for $n=k+1$

Step 4 As true for $n=3$ and if true for $n=k$, it is true for $n=k+1$. $\therefore$ true for all positive integers for $n \geq 3$

Question 3

$$(i) \frac{\sin^3 x}{\cos^3 x} + \frac{\cos^3 x}{\sin^3 x} = 0$$

$$\tan^3 x + 1 = 0$$

$$\tan^3 x = -1$$

$$\therefore \tan x = -1$$

$$(ii) \cos \theta - \sqrt{3} \sin \theta$$

$$R = \sqrt{3+1}$$

$$= 2$$

$$\tan \alpha = \sqrt{3}$$

$$\alpha = 60^\circ$$

$$= 2 \cos(\theta + 60^\circ)$$

$$2 \cos(\theta + 60^\circ) = 1$$

$$\cos(\theta + 60^\circ) = \frac{1}{2}$$

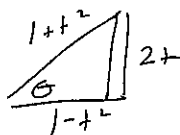
$$60^\circ \leq \theta + 60^\circ \leq 420^\circ$$

$$\theta + 60^\circ = 60^\circ, 300^\circ, 420^\circ$$

$$\theta = 0^\circ, 240^\circ, 360^\circ$$

(iii)

$$\frac{3(2t)}{1+t^2} - 4 \frac{(1-t^2)}{1+t^2} = 4$$



$$6t - 4 + 4t^2 = 4 + t + t^2$$

$$6t = 8$$

$$t = \frac{4}{3}$$

$$\tan \frac{\theta}{2} = \frac{4}{3}$$

$$\frac{\theta}{2} = 53^\circ$$

$$\theta = 106^\circ$$

Check $\theta = 180^\circ$

$$3 \sin 180^\circ - 4 \cos 180^\circ$$

$$= 0 - -4$$

$$= 4$$

$\therefore$ Solⁿs are 106° and 180°

(iv)

$$\frac{\sin 5\theta}{\sin \theta} - \frac{\cos 5\theta}{\cos \theta} = 4 \cos 2\theta$$

$$\text{LHS} = \frac{\sin 5\theta \cos \theta - \cos 5\theta \sin \theta}{\sin \theta \cos \theta}$$

$$= \frac{\sin(5\theta - \theta)}{\sin \theta \cos \theta}$$

$$= \frac{\sin 4\theta}{\sin \theta \cos \theta}$$

$$= \frac{2 \sin 2\theta \cos 2\theta}{\frac{1}{2} \sin 2\theta}$$

$$= 4 \cos 2\theta$$

$$= \text{RHS}$$

$$(iv) (a) \text{Area } \triangle ABD = \frac{1}{2} \cdot AD \cdot BD$$

$$\sin \theta = \frac{AD}{a} \quad \cos \theta = \frac{BD}{a}$$

$$a \sin \theta = AD \quad a \cos \theta = BD$$

$$\begin{aligned} \therefore A &= \frac{1}{2} a \sin \theta \cdot a \cos \theta \\ &= \frac{a^2 \sin \theta \cos \theta}{2} \end{aligned}$$

$$\begin{aligned} (b) \text{Area } \triangle ABC &= 2 \times \text{Area } \triangle ABD \\ &= a^2 \sin \theta \cos \theta \end{aligned}$$

$$\begin{aligned} \text{also, Area } \triangle ABC &= \frac{1}{2} \times a \times a \times \sin 2\theta \\ &= \frac{1}{2} a^2 \sin 2\theta \end{aligned}$$

$$\begin{aligned} \therefore \frac{1}{2} a^2 \sin 2\theta &= a^2 \sin \theta \cos \theta \\ \frac{1}{2} \sin 2\theta &= \sin \theta \cos \theta \\ \sin 2\theta &= 2 \sin \theta \cos \theta \end{aligned}$$